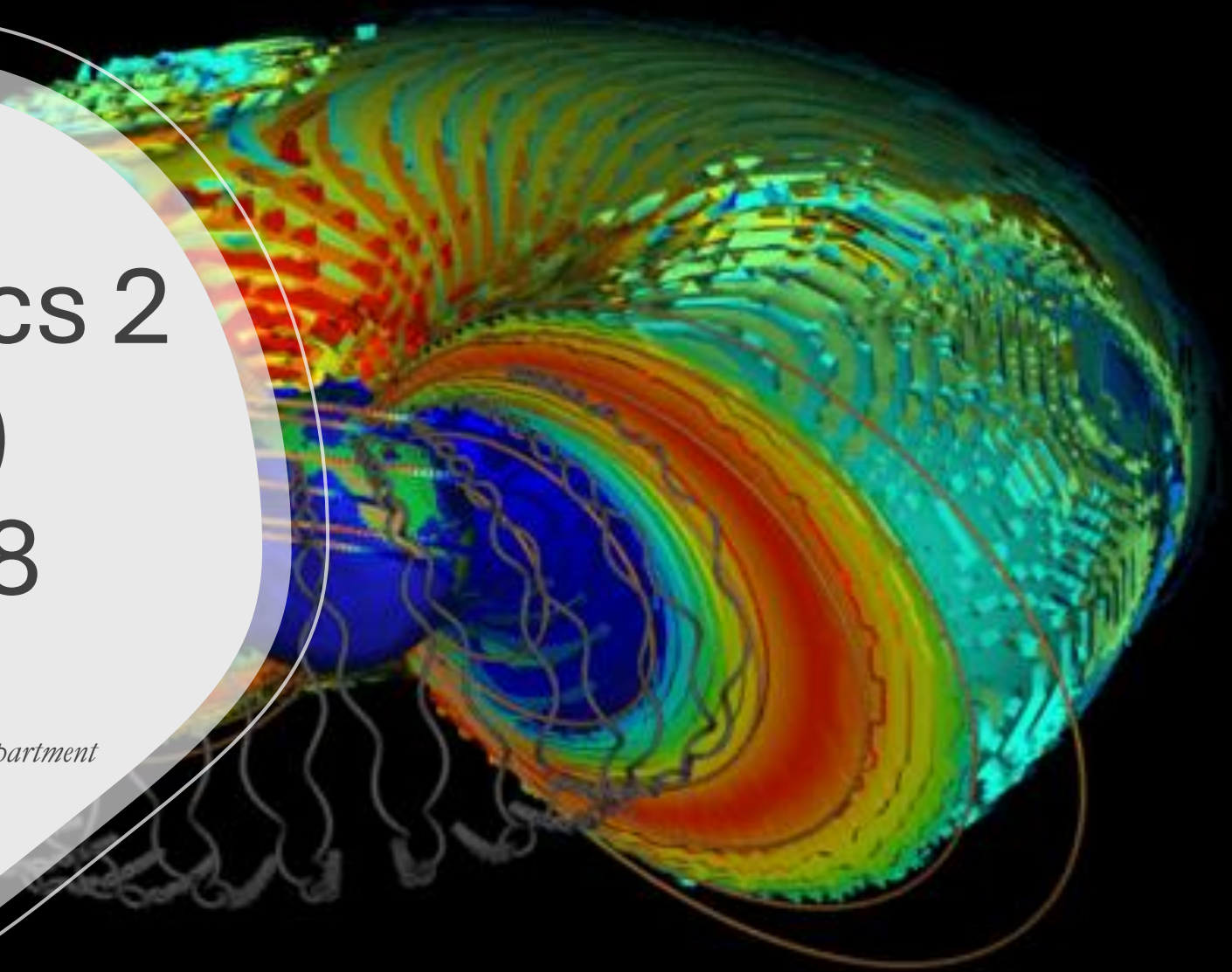


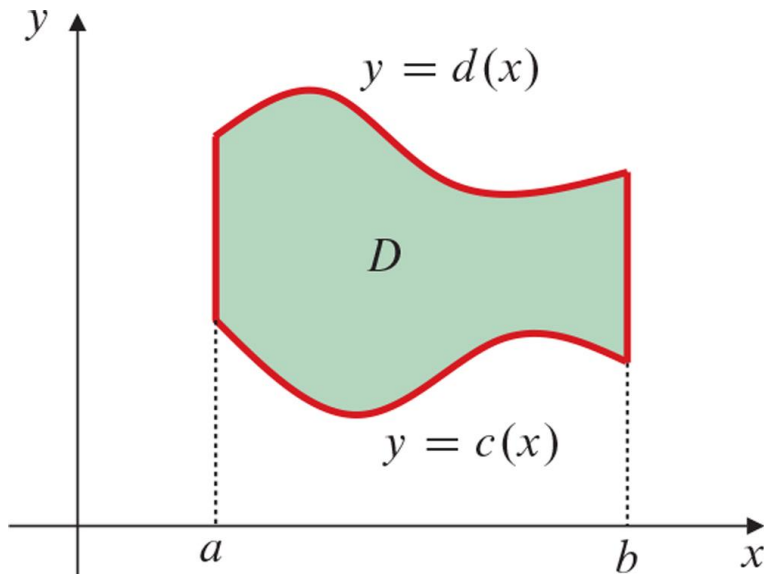
Mathematics 2 (5EZB0) Lecture 8

Koondi Mitra

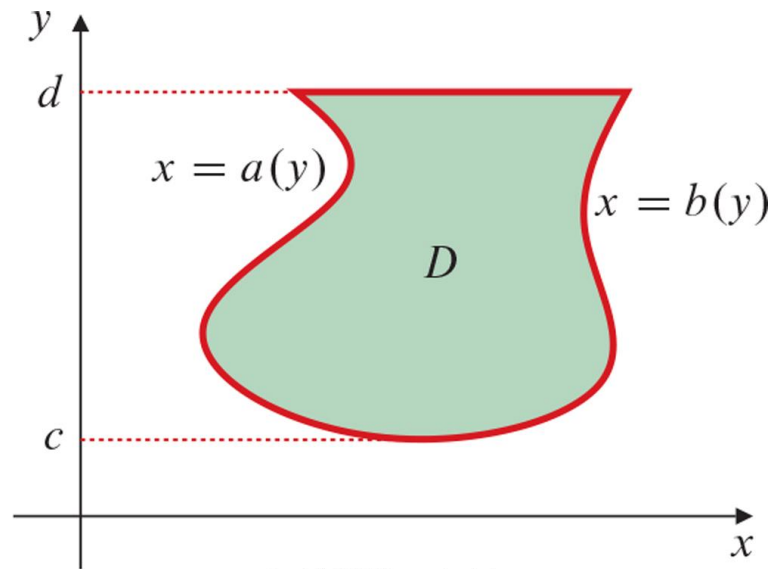
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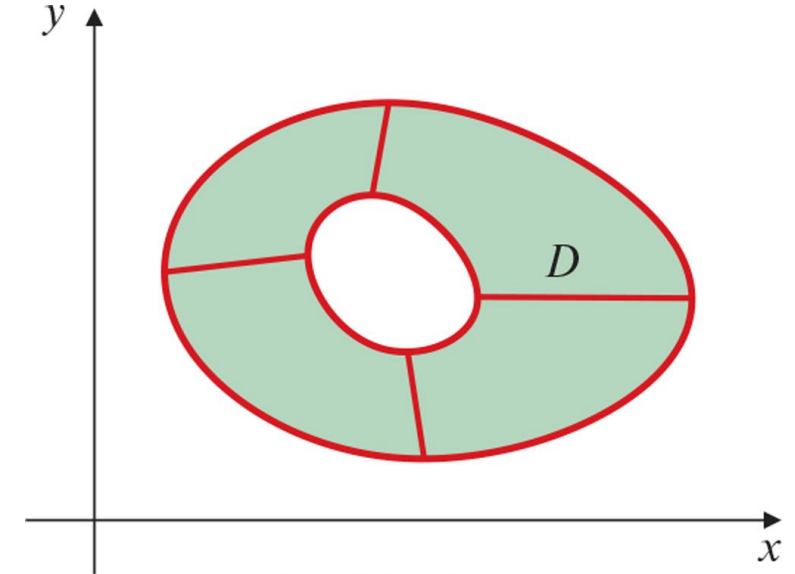
Integrals in Cartesian coordinates: domains



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x-simple domains

Let $c, d: [a, b] \rightarrow \mathbb{R}$ be continuous, and

$$\mathcal{D} := \{(x, y): a \leq x \leq b, \\ c(x) \leq y \leq d(x)\}$$

y-simple domains

Let $a, b: [c, d] \rightarrow \mathbb{R}$ be continuous, and

$$\mathcal{D} := \{(x, y): c \leq y \leq d, \\ a(y) \leq x \leq b(y)\}$$

Regular domains

Domains that can be decomposed into a finite number of x -simple or y -simple domains.

Integrals in Cartesian coordinates: computation

- *x-simple domain*: $\int \int_{\mathcal{D}} f(x, y) dA = \int_a^b \left(\int_{c(x)}^{d(x)} f(x, y) dy \right) dx$
- *y-simple domain*: $\int \int_{\mathcal{D}} f(x, y) dA = \int_c^d \left(\int_{a(y)}^{b(y)} f(x, y) dx \right) dy$

Fubini's Theorem

If $\mathcal{D}$ is both *x-simple* and *y-simple* and $f: \mathcal{D} \rightarrow \mathbb{R}$ is continuous, then both integrals are the same.

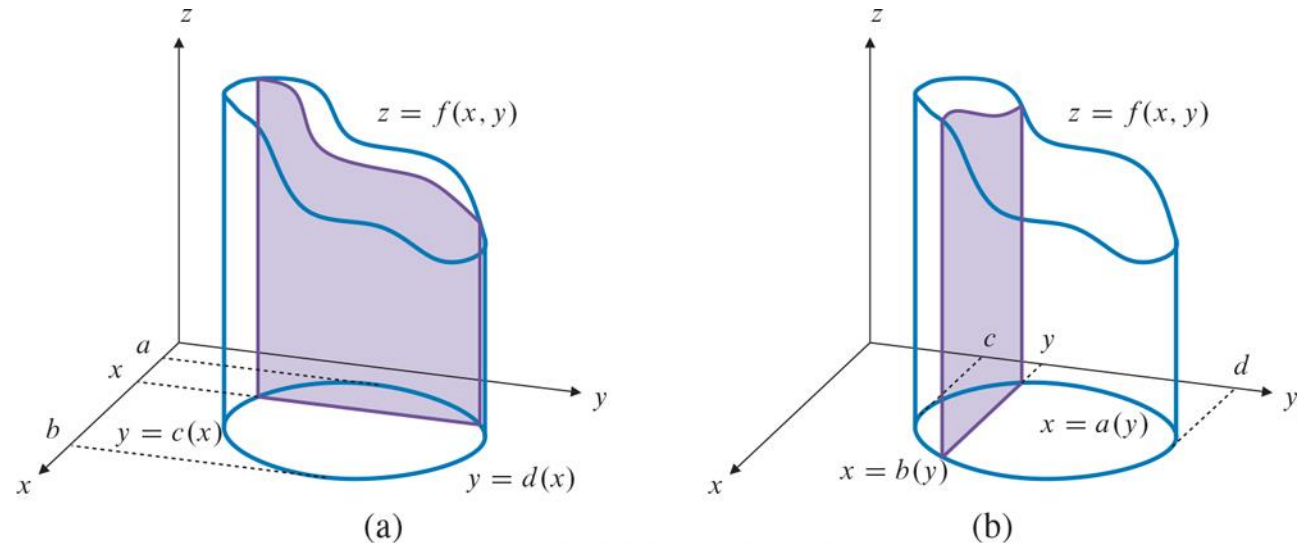
Hence, we also write (with the association $dA = dxdy$)

$$\int \int_{\mathcal{D}} f(x, y) dA \quad \text{as} \quad \int \int_{\mathcal{D}} f(x, y) dxdy$$

Question 1

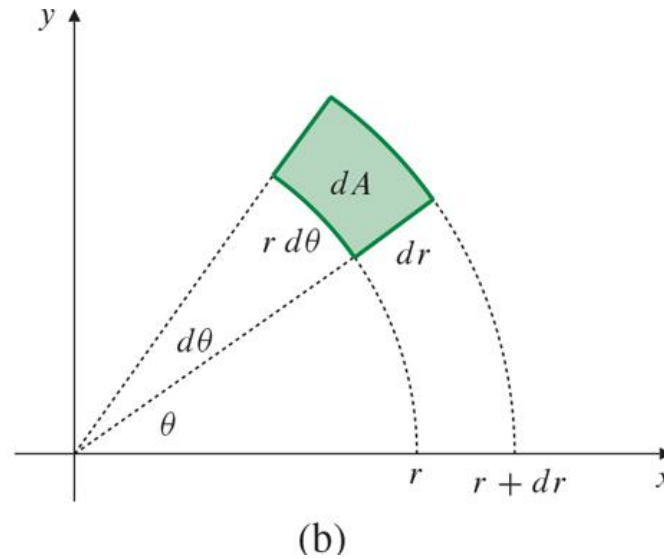
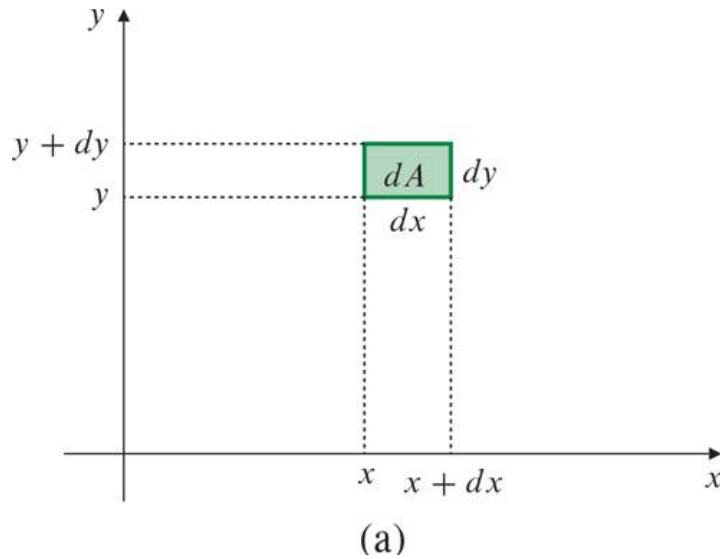
Find the integral

$$\int_0^{\pi/2} \int_y^{\pi/2} \frac{\sin x}{x} dxdy$$



Integrals in polar coordinates: domains

For functions that are easier to write in terms of cylindrical coordinates, can be integrated differently.



Let

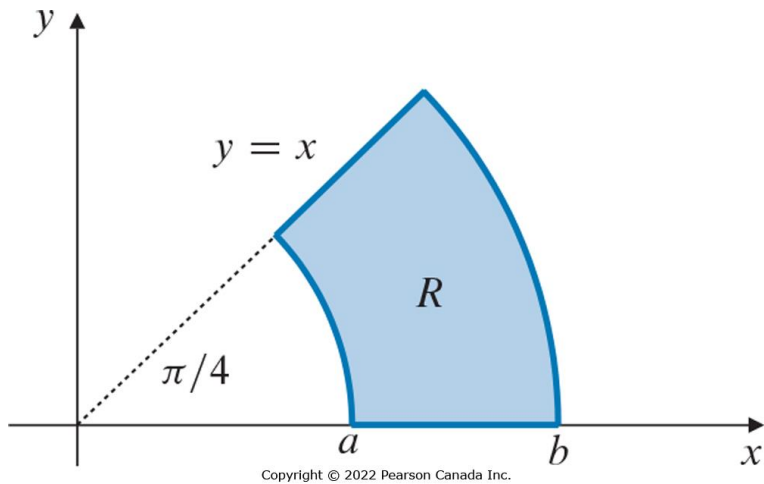
$$f(x, y) = f(r \cos \theta, r \sin \theta) = h(r, \theta)$$

Noticing that an infinitesimal dA can be represented by $dx dy = r dr d\theta$, one has

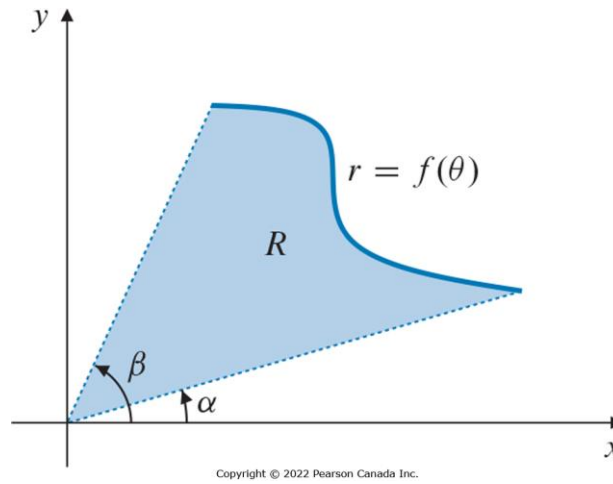
$$\begin{aligned} \iint_{\mathcal{D}} f(x, y) dA &= \iint_{\mathcal{D}} f(x, y) dx dy \\ &= \iint_{\mathcal{D}} h(r, \theta) r dr d\theta \end{aligned}$$

Integrals in polar coordinates: computing

For functions that are easier to write in terms of cylindrical coordinates, can be integrated differently.



$$\begin{aligned} & \int \int_R h(r, \theta) r dr d\theta \\ &= \int_a^b \left(\int_0^{\pi/4} h(r, \theta) d\theta \right) r dr \end{aligned}$$



$$\begin{aligned} & \int \int_R h(r, \theta) r dr d\theta \\ &= \int_\alpha^\beta \left(\int_0^{f(\theta)} h(r, \theta) r dr \right) d\theta \end{aligned}$$

Let

$$f(x, y) = f(r \cos \theta, r \sin \theta) = h(r, \theta)$$

Noticing that an infinitesimal dA can be represented by $dx dy = r dr d\theta$, one has

$$\begin{aligned} \int \int_{\mathcal{D}} f(x, y) dA &= \int \int_{\mathcal{D}} f(x, y) dx dy \\ &= \int \int_{\mathcal{D}} h(r, \theta) r dr d\theta \end{aligned}$$

Questions 2.1

Prove that $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$

Questions 2.2

For $\mathcal{D} := \{x \leq y, 1 \leq x^2 + y^2 \leq 2\}$, find $\int \int_{\mathcal{D}} xy dx dy$

Change of variables: domains

If a variable transformation is made from $(x, y) \rightarrow (u, v)$ then how does the integral change?

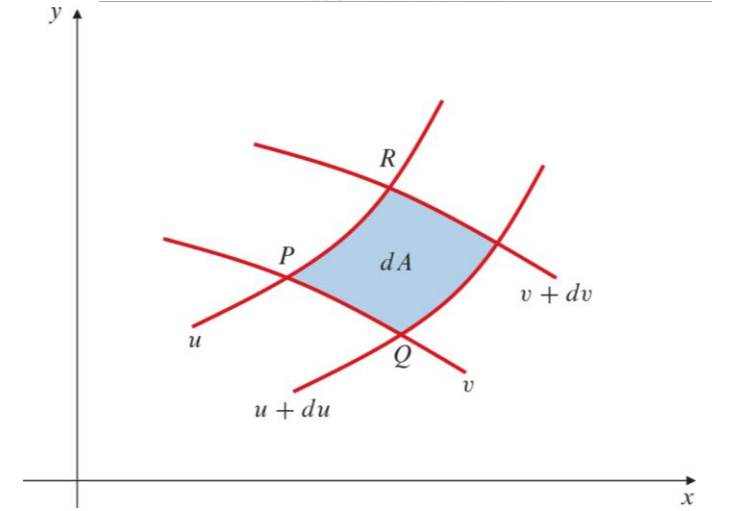
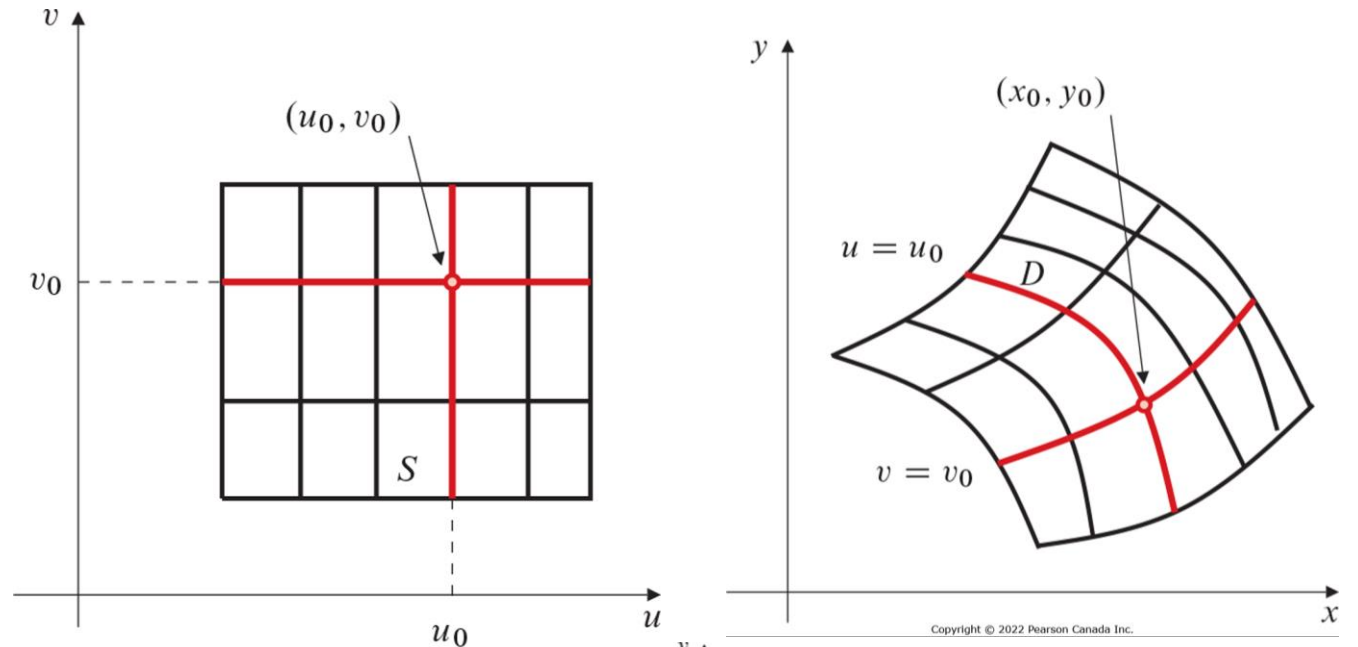
Let

$$\begin{aligned}x &= x(u, v), \\y &= y(u, v).\end{aligned}$$

Then formally, the infinitesimal area dA transforms as

$$dA = dxdy = \left| \det \left(\frac{\partial(x, y)}{\partial(u, v)} \right) \right| dudv$$

Let the domain of integral $\mathcal{D}$ in (x, y) plane be changed to S in (u, v) plane



Change of variables: integrals

Let

$$x = x(u, v)$$

$$y = y(u, v)$$

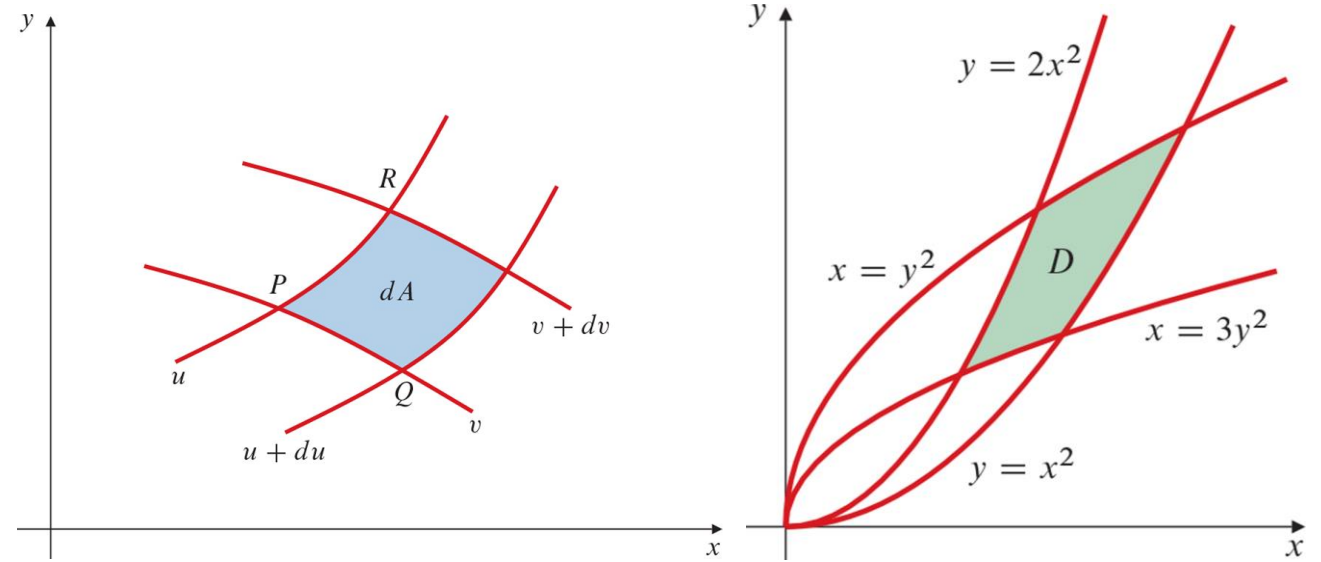
and

$$f(x(u, v), y(u, v)) = h(u, v).$$

Let the domain of integral $\mathcal{D}$ in (x, y) plane be changed to S in (u, v) plane

$$\int \int_{\mathcal{D}} f(x, y) dx dy = \int \int_S h(u, v) \left| \det \left(\frac{\partial(x, y)}{\partial(u, v)} \right) \right| du dv$$

Questions 3.1 Show that in polar coordinate,
 $dA = r dr d\theta$



Questions 3.2

Using coordinate transform, find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$

Questions 3.3

Find the integral $\int \int_{\mathcal{D}} xy dx dy$ for the domain $\mathcal{D}$ shown in the picture

Triple integrals

$$\iiint_{\mathcal{D}} f(x, y, z) dx dy dz$$

- Volume of domain $\mathcal{D}$: $\int \int \int_{\mathcal{D}} dx dy dz$
- Mass of solid: $\int \int \int_{\mathcal{D}} \rho(x, y, z) dx dy dz$
- For $\mathcal{D} := \{(x, y, z): a \leq x \leq b, c(x) \leq y \leq d(x), e(x, y) \leq z \leq f(x, y)\}$

$$\int \int \int_{\mathcal{D}} f(x, y, z) dx dy dz = \int_a^b \left(\int_{c(x)}^{d(x)} \left(\int_{e(x, y)}^{f(x, y)} f(x, y, z) dz \right) dy \right) dx$$

- If f is continuous, and an integral is well defined in two separate orders, then the order of integration can be exchanged
- Higher order integrations are also written simply as $\int_{\mathcal{D}} f dx$



$$I = \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} y dz.$$

$$I = \int_0^1 dx \int_0^{1-x} dz \int_0^{1-x-z} y dy,$$

$$I = \int_0^1 dy \int_0^{1-y} dx \int_0^{1-x-y} y dz,$$

$$I = \int_0^1 dy \int_0^{1-y} dz \int_0^{1-y-z} y dx,$$

$$I = \int_0^1 dz \int_0^{1-z} dx \int_0^{1-x-z} y dy,$$

$$I = \int_0^1 dz \int_0^{1-z} dy \int_0^{1-y-z} y dx.$$

Triple integrals

$$\iiint_{\mathcal{D}} f(x, y, z) dx dy dz$$

➤ Volume of domain $\mathcal{D}$: $\int \int \int_{\mathcal{D}} dx dy dz$

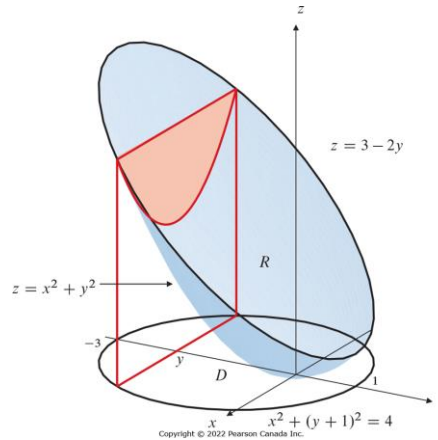
➤ Mass of solid: $\int \int \int_{\mathcal{D}} \rho(x, y, z) dx dy dz$

➤ For $\mathcal{D} := \{(x, y, z): a \leq x \leq b, c(x) \leq y \leq d(x), e(x, y) \leq z \leq f(x, y)\}$

$$\int \int \int_{\mathcal{D}} f(x, y, z) dx dy dz = \int_a^b \left(\int_{c(x)}^{d(x)} \left(\int_{e(x,y)}^{f(x,y)} f(x, y, z) dz \right) dy \right) dx$$

➤ If f is continuous, and an integral is well defined in two separate orders, then the order of integration can be exchanged

➤ Higher order integrations are also written simply as $\int_{\mathcal{D}} f dx$



Questions 4.1

What is the z -coordinate of the center of mass of a uniformly dense tetrahedron with vertices at $(\pm 1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$

$$\bar{x}_i = \left(\int_{\mathcal{D}} x_i \rho(x) dx \right) / \left(\int_{\mathcal{D}} \rho(x) dx \right)$$

Questions 4.2

What is the moment of inertia of the sphere $x^2 + y^2 + z^2 = R^2$ along the z -axis?

$$I_z = \int_{\mathcal{D}} r^2 \rho(x) dx$$

Questions 4.3

Find the volume of the region under the plane $z = 3 - 2y$ and above the paraboloid $z = x^2 + y^2$

Change of variables

Now consider the change of variable

$$x = x(u, v, w)$$

$$y = y(u, v, w)$$

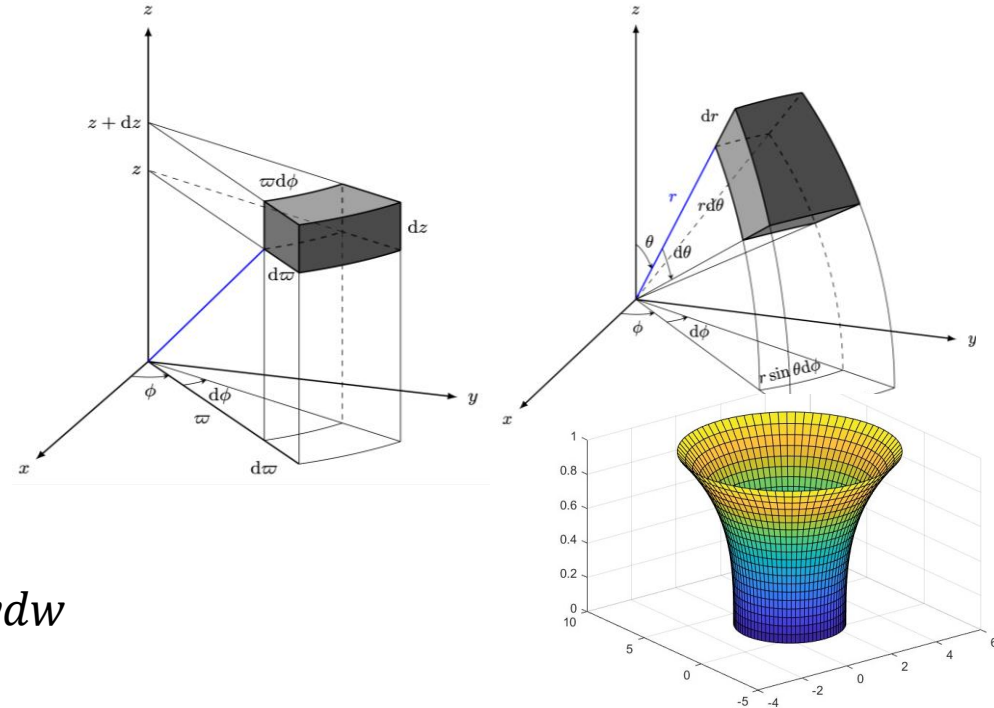
$$z = z(u, v, w)$$

➤ Let $f(x(u, v, w), y(u, v, w), z(u, v, w)) = h(u, v, w)$

➤ Let the domain $\mathcal{D}$ be transformed into S in (u, v, w) coordinates

Then, the area element changes by $dx dy dz = \left| \det \left(\frac{\partial(x, y, z)}{\partial(u, v, w)} \right) \right| du dv dw$

$$\iiint_{\mathcal{D}} f(x, y, z) dx dy dz = \iiint_S h(u, v, w) \left| \det \left(\frac{\partial(x, y, z)}{\partial(u, v, w)} \right) \right| du dv dw$$



Questions 5

Find the volume of the curve
between $z = 0$ and 1

$$\frac{(x-1)^2}{a^2} + \frac{(y-2)^2}{b^2} = z^3 + 1$$